DREAM-High: From Genes to Patients

Teaching Cancer Systems Biology Across Scales

Diana Murray[1], Claudia Ludwig[2], Andrew Gentles[3], Jason Somarelli[4], Angie Bowen[5]

[1]Columbia University · [2]Institute for Systems Biology · [3]Stanford University · [4]Duke University · [5]Sage Bionetworks

Authentic cancer data. Student-led investigation. A growing scientific community.

120 students have participated to date

5 alumni helped teach and expand the 2026 course

$75,000 additional grant for proposed 2027 developments

WHY DREAM-HIGH?

High school students investigate cancer across biological scales, connecting molecular variation with cellular behavior and clinical outcomes while developing computational skills.

FROM GENES TO PATIENTS

01 Genes & tumors

Integrate TCGA gene expression and clinical data; identify tumor subtypes and explore expression patterns with heat maps.

02 Cells & microenvironments

Compare cancer cell line expression profiles with physical phenotypes measured in environments that mimic the extracellular environment.

03 Patients & outcomes

Use METABRIC data and linear regression to explore associations between cancer-associated gene expression and patient survival.

HOW STUDENTS LEARN

An online course combines browser-based RStudio, Pathwright activities, mentor-supported breakout rooms, and capstone projects.

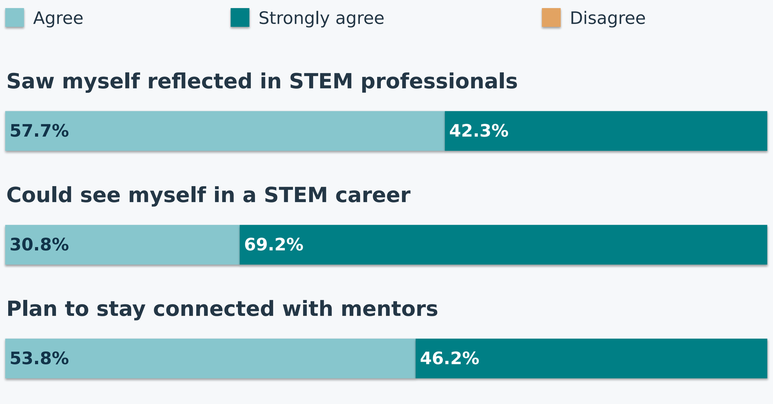
A question of their own

Students choose a question, analyze authentic data, interpret findings, and communicate their work through a project website. Scientists and a patient advocate connect the analyses to research careers and lived experience.

LEARNING & COMMUNITY

End-of-program reflections reveal strong student-reported connection to science and mentors.

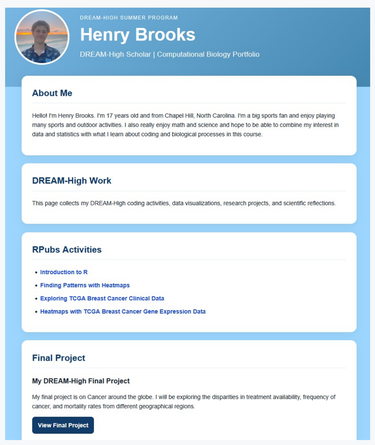
Responses after DREAM-High



Four-point scale: strongly disagree, disagree, agree, strongly agree. Strongly disagree: 0% for all four items. Percentages from supplied slides; respondent count not provided.

These reflections describe perceptions and intentions; they do not directly measure coding proficiency.

STUDENT WORK IN PRACTICE



From analysis to communication

Henry Brooks' portfolio connects R activities, data visualizations, and a final project exploring geographic differences in cancer and treatment access.

Example student portfolio from the supplied course slides.

PARTICIPANTS → CONTRIBUTORS

Five alumni helped teach and expand the 2026 course through breakout-room mentoring, learning activities, and final-project support.

Students from the 2026 cohort want to help plan, teach, and mentor in 2027—creating a pathway from participation to leadership.

PROPOSED 2027 DEVELOPMENTS

The additional $75,000 grant creates an opportunity to invest in people, curriculum, and creative collaboration.

More independent R coding

Respond to student feedback: modify code → complete steps → debug scripts → design an analysis. Offer core tasks and optional challenges.

More voices in cancer research

Expand small-group conversations with scientists and patient advocates; connect analysis with research questions and patient experience.

Alumni planning & mentoring

Form a student advisory group; train returning students for supported teaching and peer-mentor roles, with stipends where permitted.

DREAM-High Artist-in-Residence

Lead "From Data to Story": students translate findings into illustrations, visual narratives, or animations with scientific review and patient input.

Evaluate growth with coding and interpretation tasks alongside reflections and alumni follow-up.

JOIN THE DREAM-HIGH COMMUNITY

We welcome CSBC partners to contribute research-based modules, meet students, support mentors, and broaden participation.

LET'S CONNECT

Diana Murray
dm527@cumc.columbia.edu